Formula Sheet

Final Examination

Subject Title: Derivative Securities

Subject Code: FNCE30007

Cost of Asset/Price Realised $= F_0 + (S_1 - F_1) = F_0 + Basis_1$

$F_0 + (S^*_1 - F_1) + (S_1 - S^*_1)$

$N^* = \beta \frac{V_A}{V_F}$

$N^* = (\beta - \beta^*) \frac{V_A}{V_F}$

$h^* = \frac{\sigma_{sf}}{\sigma_f^2} = \rho_{sf} \frac{\sigma_s}{\sigma_f}$

$N^* = h^* \frac{V_A}{V_F}$

$r_c = m \cdot ln\left(1 + \frac{r_m}{m}\right)$

$r^m = m \times (e^{\frac{r^*}{m}} - 1)$

$F_0 = S_0 e^{rT}$

$F_0 = (S_0 - I)e^{rT}$

$F_0 = S_0 e^{(r-q)T}$

$F_0 = S_0 e^{(r-r_f)T}$

$F_0 e^{yT} = (S_0 + U)e^{rT}$

$F_0 = S_0 e^{(r+u-y)T}$

$f = (F - K)e^{-rT}$

$f = (K - F)e^{-rT}$

$f = F - K$

$f = K - F$

$c \leq S_0$

$C \leq S_0$

$p \leq Ke^{-rT}$

$P \leq K$

$c \geq max(S_0 - Ke^{-rT}, 0)$

$p \geq max(Ke^{-rT} - S_0, 0)$

$c + Ke^{-rT} = p + S_0$

$c \geq \max(S_0 - D - Ke^{-rT}, 0)$

$p \geq \max(D + Ke^{-rT} - S_0, 0)$

$C \geq \max(S_0 - Ke^{-rT}, 0)$

$P \geq \max(K - S_0, 0)$

$S_0 - K < C - P < S_0 - Ke^{-rT}$

$c + D + Ke^{-rT} = p + S_0$

$S_0 - D - K < C - P < S_0 - Ke^{-rT}$

$\Delta = \frac{f_u - f_d}{uS - dS}$

$f = \frac{pf_u + (1-p)f_d}{e^{r\Delta t}}$

$p = \frac{e^{r\Delta t} - d}{u - d}$

$f = \frac{p^2 f_{uu} + 2p(1-p)f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}$

$f = \frac{\left(p^3 f_{uuu} + 3p^2(1-p)f_{uud} + 3p(1-p)^2 f_{udd} + (1-p)^3 f_{ddd}\right)}{e^{3 \times r \times \Delta t}}$

$\pi = \Delta u S_o - f_u = \Delta d S_o - f_d$

$u = e^{\sigma\sqrt{\Delta t}}$

$d = e^{-\sigma\sqrt{\Delta t}} = \frac{1}{u}$

$c + Ke^{-rT} = p + F_0 e^{-rT}$

$F_0 e^{-rT} - K < C - P < F_0 - Ke^{-rT}$

$c \geq (F_0 - K)e^{-rT}$

$p \geq (K - F_0)e^{-rT}$

$C \geq F_0 - K$

$P \geq K - F_0$

$\Delta = \frac{f_u - f_d}{F_0 u - F_0 d}$

$p = \frac{1-d}{u-d}$

$lnS_T \sim \varphi\left[ln\, S_0 + \left(\mu - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right]$

$E(S_T) = S_0 e^{\mu T}$

$var(S_T) = S_0^2 e^{2\mu T}\left(e^{\sigma^2 T} - 1\right)$

$Z = \frac{X - u}{\sigma}$

$c = S_0 N(d_1) - Ke^{-rT} N(d_2)$

$p = Ke^{-rT} N(-d_2) - S_0 N(-d_1)$

$d_1 = \frac{ln\left(\frac{S_0}{K}\right) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}$

$d_2 = \frac{ln\left(\frac{S_0}{K}\right) + (r - \sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$

$$c = e^{-rT}[F_0 N(d_1) - KN(d_2)]$$

$$p = e^{-rT}[KN(-d_2) - F_0 N(-d_1)]$$

$$d_1 = \frac{\ln(F_0/K)+\sigma^2 T/2}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(F_0/K)-\sigma^2 T/2}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$c \geq S_0 e^{-qT} - Ke^{-rT}$$

$$p \geq Ke^{-rT} - S_0 e^{-qT}$$

$$c + Ke^{-rT} = p + S_0 e^{-qT}$$

$$c + Ke^{-rT} = p + S_0 e^{-r_f T}$$

$$p = \frac{e^{(r-q)\Delta t}-d}{u-d}$$

$$c = S_0 e^{-qT} N(d_1) - Ke^{-rT} N(d_2)$$

$$p = Ke^{-rT}\, N(-d_2) - S_0 e^{-qT} N(-d_1)$$

$$d_1 = \frac{\ln(S_0/K)+(r-q+\sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K)+(r-q-\sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$c = S_0 e^{-r_f T} N(d_1) - Ke^{-rT} N(d_2)$$

$$p = Ke^{-rT}\, N(-d_2) - S_0 e^{-r_f T} N(-d_1)$$

$$d_1 = \frac{\ln(S_0/K)+(r-r_f+\sigma^2/2)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln(S_0/K)+(r-r_f-\sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$c = [S_0 - PV(D)]N(d_1) - Ke^{-rT} N(d_2)$$

$$p = Ke^{-rT} N(-d_2) - [S_0 - PV(D)]N(-d_1)$$

$$d_1 = \frac{\ln\left[\frac{S_0 - PV(D)}{K}\right]+\left(r+\frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}$$

$$d_2 = \frac{\ln[(S_0 - PV(D))/K]+(r-\sigma^2/2)T}{\sigma\sqrt{T}} = d_1 - \sigma\sqrt{T}$$

$$Number\ of\ contracts = \frac{Value\ to\ be\ hedged}{Value\ one\ put\ option\ will\ hedge}$$

$$\Delta\ of\ long\ call = N(d_1)$$

$$\Delta\ of\ short\ call = -N(d_1)$$

$$\Delta\ of\ long\ put = N(d_1) - 1$$

$$\Delta\ of\ short\ put = 1 - N(d_1)$$

$$\Delta\ of\ long\ call = e^{-qT} N(d_1)$$

$$\Delta\ of\ short\ call = -e^{-qT} N(d_1)$$

$$\Delta\ of\ long\ put = e^{-qT}(N(d_1) - 1)$$

$$\Delta\ of\ short\ put = e^{-qT}(1 - N(d_1))$$

$$\Delta\ of\ long\ call = e^{-r_f T} N(d_1)$$

$\Delta\ of\ short\ call = -e^{-r_f T} N(d_1)$

$\Delta\ of\ long\ put = e^{-r_f T}(N(d_1) - 1)$

$\Delta\ of\ short\ put = e^{-r_f T}(1 - N(d_1))$

$\Delta\ of\ long\ call = e^{-rT} N(d_1)$

$\Delta\ of\ short\ call = -e^{-rT} N(d_1)$

$\Delta\ of\ long\ put = e^{-rT}(N(d_1) - 1)$

$\Delta\ of\ short\ put = e^{-rT}(1 - N(d_1))$

$\Delta of long forward = 1$

$\Delta of short forward = -1$

$\Delta\ of\ long\ forward = e^{-qT}$

$\Delta\ of\ short\ forward = -e^{-qT}$

$\Delta\ of\ long\ forward = e^{-r_f T}$

$\Delta\ of\ short\ forward = -e^{-r_f T}$

$\Delta\ of\ long\ futures = e^{rT}$

$\Delta\ of\ short\ futures = -e^{rT}$

$\Delta\ of\ long\ futures = e^{(r-q)T}$

$\Delta\ of\ short\ futures = -e^{(r-q)T}$

$\Delta\ of\ long\ futures = e^{(r-r_f)T}$

$\Delta\ of\ short\ futures = -e^{(r-r_f)T}$

$P = \sum c * e^{-R_i t_i} + FV * e^{-R_n t_n}$

$RF = (R_2 T_2 - R_1 T_1)/(T_2 - T_1)$

Table for $N(x)$ When $x \leq 0$

No Interpolation is required for the calculation of $N(d_1)$ and $N(d_2)$

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
–0.0	0.5000	0.4960	0.4920	0.4880	0.4840	0.4801	0.4761	0.4721	0.4681	0.4641
–0.1	0.4602	0.4562	0.4522	0.4483	0.4443	0.4404	0.4364	0.4325	0.4286	0.4247
–0.2	0.4207	0.4168	0.4129	0.4090	0.4052	0.4013	0.3974	0.3936	0.3897	0.3859
–0.3	0.3821	0.3783	0.3745	0.3707	0.3669	0.3632	0.3594	0.3557	0.3520	0.3483
–0.4	0.3446	0.3409	0.3372	0.3336	0.3300	0.3264	0.3228	0.3192	0.3156	0.3121
–0.5	0.3085	0.3050	0.3015	0.2981	0.2946	0.2912	0.2877	0.2843	0.2810	0.2776
–0.6	0.2743	0.2709	0.2676	0.2643	0.2611	0.2578	0.2546	0.2514	0.2483	0.2451
–0.7	0.2420	0.2389	0.2358	0.2327	0.2296	0.2266	0.2236	0.2206	0.2177	0.2148
–0.8	0.2119	0.2090	0.2061	0.2033	0.2005	0.1977	0.1949	0.1922	0.1894	0.1867
–0.9	0.1841	0.1814	0.1788	0.1762	0.1736	0.1711	0.1685	0.1660	0.1635	0.1611
–1.0	0.1587	0.1562	0.1539	0.1515	0.1492	0.1469	0.1446	0.1423	0.1401	0.1379
–1.1	0.1357	0.1335	0.1314	0.1292	0.1271	0.1251	0.1230	0.1210	0.1190	0.1170
–1.2	0.1151	0.1131	0.1112	0.1093	0.1075	0.1056	0.1038	0.1020	0.1003	0.0985
–1.3	0.0968	0.0951	0.0934	0.0918	0.0901	0.0885	0.0869	0.0853	0.0838	0.0823
–1.4	0.0808	0.0793	0.0778	0.0764	0.0749	0.0735	0.0721	0.0708	0.0694	0.0681
–1.5	0.0668	0.0655	0.0643	0.0630	0.0618	0.0606	0.0594	0.0582	0.0571	0.0559
–1.6	0.0548	0.0537	0.0526	0.0516	0.0505	0.0495	0.0485	0.0475	0.0465	0.0455
–1.7	0.0446	0.0436	0.0427	0.0418	0.0409	0.0401	0.0392	0.0384	0.0375	0.0367
–1.8	0.0359	0.0351	0.0344	0.0336	0.0329	0.0322	0.0314	0.0307	0.0301	0.0294
–1.9	0.0287	0.0281	0.0274	0.0268	0.0262	0.0256	0.0250	0.0244	0.0239	0.0233
–2.0	0.0228	0.0222	0.0217	0.0212	0.0207	0.0202	0.0197	0.0192	0.0188	0.0183
–2.1	0.0179	0.0174	0.0170	0.0166	0.0162	0.0158	0.0154	0.0150	0.0146	0.0143
–2.2	0.0139	0.0136	0.0132	0.0129	0.0125	0.0122	0.0119	0.0116	0.0113	0.0110
–2.3	0.0107	0.0104	0.0102	0.0099	0.0096	0.0094	0.0091	0.0089	0.0087	0.0084
–2.4	0.0082	0.0080	0.0078	0.0075	0.0073	0.0071	0.0069	0.0068	0.0066	0.0064
–2.5	0.0062	0.0060	0.0059	0.0057	0.0055	0.0054	0.0052	0.0051	0.0049	0.0048
–2.6	0.0047	0.0045	0.0044	0.0043	0.0041	0.0040	0.0039	0.0038	0.0037	0.0036
–2.7	0.0035	0.0034	0.0033	0.0032	0.0031	0.0030	0.0029	0.0028	0.0027	0.0026
–2.8	0.0026	0.0025	0.0024	0.0023	0.0023	0.0022	0.0021	0.0021	0.0020	0.0019
–2.9	0.0019	0.0018	0.0018	0.0017	0.0016	0.0016	0.0015	0.0015	0.0014	0.0014
–3.0	0.0014	0.0013	0.0013	0.0012	0.0012	0.0011	0.0011	0.0011	0.0010	0.0010
–3.1	0.0010	0.0009	0.0009	0.0009	0.0008	0.0008	0.0008	0.0008	0.0007	0.0007
–3.2	0.0007	0.0007	0.0006	0.0006	0.0006	0.0006	0.0006	0.0005	0.0005	0.0005
–3.3	0.0005	0.0005	0.0005	0.0004	0.0004	0.0004	0.0004	0.0004	0.0004	0.0003
–3.4	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003	0.0002
–3.5	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002
–3.6	0.0002	0.0002	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
–3.7	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
–3.8	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001
–3.9	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
–4.0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table for $N(x)$ When $x \geq 0$

No Interpolation is required for the calculation of N(d1) and N(d2)

x	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9986	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998
3.5	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998	0.9998
3.6	0.9998	0.9998	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.7	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.8	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
3.9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4.0	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000